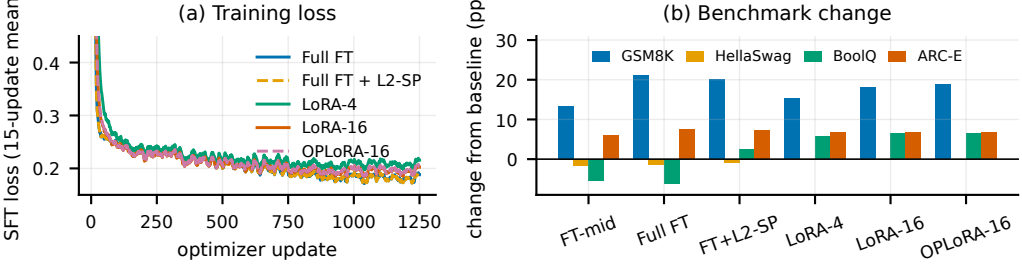
(a) Training loss
SFT loss (15-update mean)
0.4
0.3
0.2
0
250
500
750
1000
1250
optimizer update
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(b) Benchmark change
change from baseline (pp)
30
20
10
0
GSM8K
HellaSwag
BoolQ
ARC-E
FT-mid
Full FT
FT+L2-SP
LoRA-4
LoRA-16
OPLoRA-16